

Aim Determination of transformer equivalent circuit from open circuit and short circuit test.

Theory a) Open Circuit Test $\Rightarrow$ The purpose is to determine no-load current and losses of the transformer because of which no-load parameters are defined. This test is performed on the primary winding of the transformer. The wattmeter, ammeter and voltmeter are connected to their primary winding. The nominal rated voltage is supplied to their primary winding with the help of AC source.

W_0 = wattmeter reading V_1 = Voltmeter reading I_0 = Ammeter

Then the iron loss of transformer $P_i = W_0$

$$W_0 = (V_1)(I_0)\cos\phi \quad \text{--- ①}$$

The no load power factor is

$$\cos\phi = W_0/V_1 I_0$$

Working component is

$$I_w = W_0/V_1 \quad \text{--- ②}$$

Putting value of W_0 from eq ① in eq ②

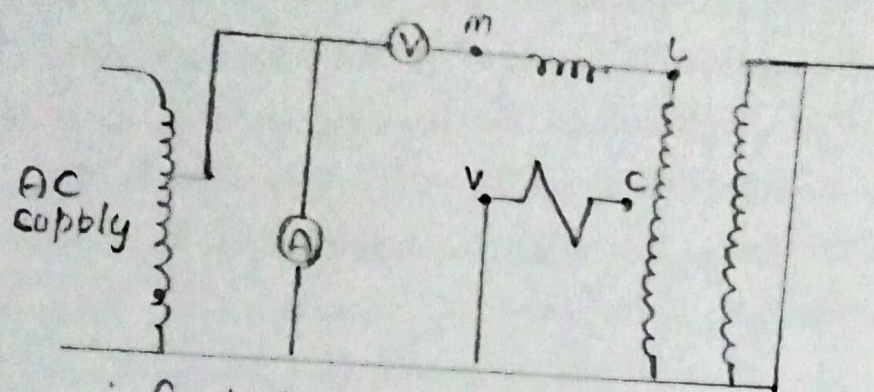
$$I_w = I_0 \cos\phi$$

magnetizing component is

$$I_m = (I_0^2 - I_w^2)^{1/2}$$

Equivalent exciting resistance $R_0 = V_1/I_w$

$$R_0 = V_1/I_m$$



∴ Equivalent circuit diagram for open circuit test .

- b) Short Circuit Test $\Rightarrow$ The short circuit test is performed for determining the below mention parameters of the transformer :-
- i) It determines the copper loss occurs on the full load. The copper loss is used for finding the efficiency of the transformer.
 - ii) The equivalent resistance, impedance, and leakage reactance are known by the short circuit test.

Let,

W_c = Wattmeter Reading V_{sc} = Voltmeter Reading I_{sc} = Ammeter Reading
Then the full load copper loss of the transformer is given by

$$P_c = [I_{sc}/\sqrt{2}]^2 = W_c$$

$$I_{sc}^2 R_s = W_c$$

Equivalent Resistance referred, $R_s = W_c / (I_{sc}^2)$

Equivalent Impedance referred to secondary side

$$Z_s = V_{sc} / I_{sc}$$

The equivalent reactance referred to secondary side

$$X_s = [(Z_s^2 - R_s^2)]^{1/2}$$

Procedure: 1) Make a proper connection by connecting the nodes.

2) Then click check button to check connection is correct or not.

3) Then click autotransformer to get the change in all meter.

4) Then click add to table button to get the reading of open open circuit transformer.

5) Then we have done open circuit transformer test. Then click short button for short circuit test.

6) Make the proper connection by clicking the arrow or by connecting.

7) Then click check button to check connection is correct or not.

8) Then we have done short circuit transformer test. We have result in diagram form.

MCB



[Instruction](#)

Check Connection

Reset

Add to Table



Autotransformer



P1

P2

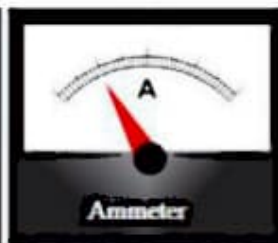
Transformer



Poc (Watt)	Ioc (A)	Voc (V)
50	0.9	230

Short circuit test

MCB



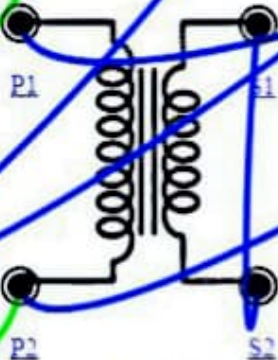
Check Connection

Reset

[Instruction](#)

Add to Table

Autotransformer



Transformer

Psc (Watt)

Isc (A)

Vsc (Volt)

37.5

4.5

11

Submit

